



CPSC 100

Computational Thinking

Algorithm, Classifiers and Trees!

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia



Agenda

- Course Admin
- Debrief on Moore's Law
- Recap Classification
- Decision Trees

Course Admin

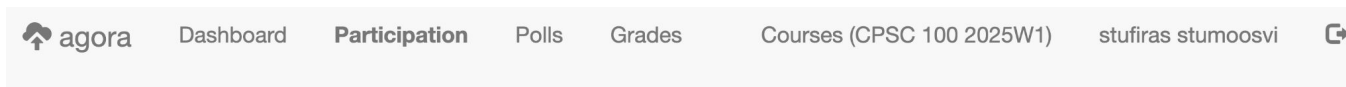
Debrief on Moore's Law



Agora: Discussions and Polling

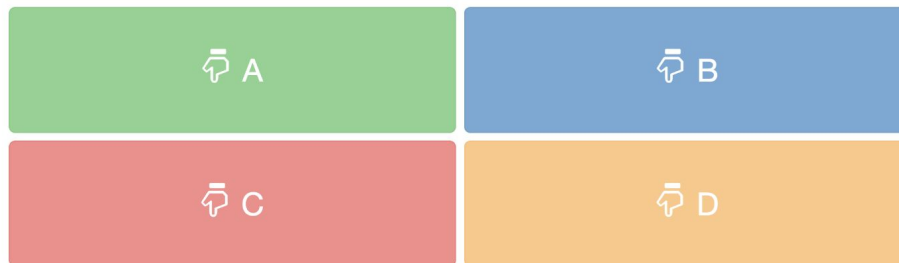
Page: agora.students.cs.ubc.ca (login with your CWL)

Enroll code: **conformitygate**



CPSC 100 2025W1

Participation Points: + (Phantom)



Message Board

Lecture not started.



Participation Question

Is Moore's Law still valid in 2025?

Yes; explain

No; explain

-

-



Soccer League:

Do we cancel the game?



Soccer League: Cancel Game?

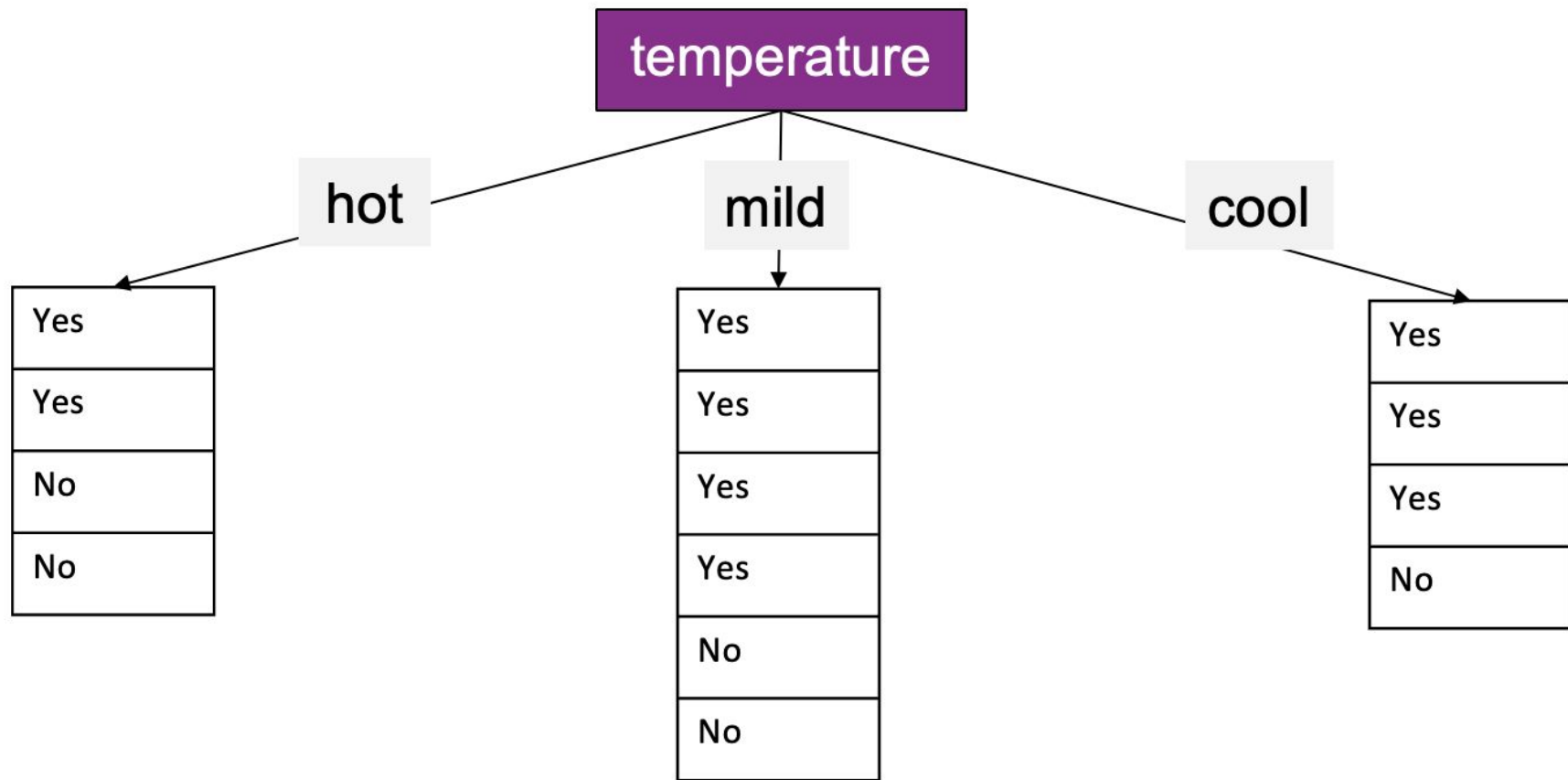
- Build a **decision tree** to help officials decide
- Assume that decisions are the same given the same information
- The leaf nodes should be whether or not to play
- The non-leaf nodes should be **attributes** (e.g., Outlook, Windy)
- The edges should be **conditions** (e.g., sunny, hot, normal)

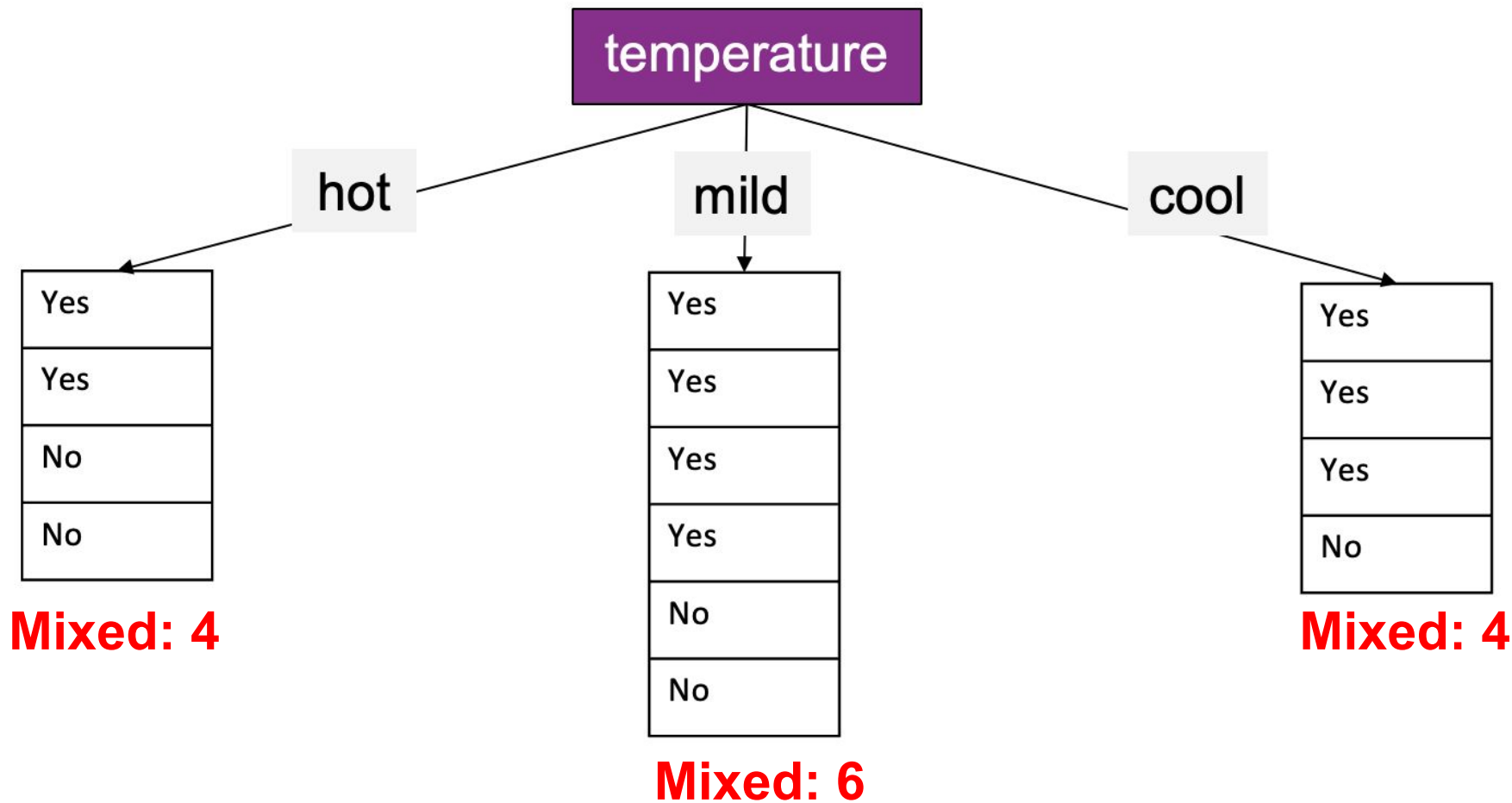
Want to have as few mixed “Yes” and “No” answers together in groups as possible.

At the start we have 14 mixed Yes’s/No’s

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

**What happens if
we split data on
Temperature?**





What is the
uncertainty
(entropy) in our
data?

Overall entropy = 4 + 4 + 6 = 14

temperature

hot

mild

cool

Yes
Yes
No
No

Mixed: 4

Yes
Yes
Yes
Yes
No
No

Mixed: 6

Yes
Yes
Yes
No

Mixed: 4

Overall entropy = 4 + 4 + 6 = 14

temperature

hot

mild

cool

Yes
Yes
No
No

Mixed: 4

Yes
Yes
Yes
Yes
No
No

Mixed: 6

Note: The entropy for cool would be 0 if all of them were Yes or all of them were No (we are using simple math)

Yes
Yes
Yes
No

Mixed: 4

In-class Activity

[Groups of 3-4]

What's the
entropy if you
split on the
Outlook?

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Clicker Question



Q: What's the entropy if you split on Outlook?

A. 0

B. 5

C. 10

D. 14

E. None of the above

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Clicker Question



Q: What's the entropy if you split on Outlook?

A. 0

B. 5

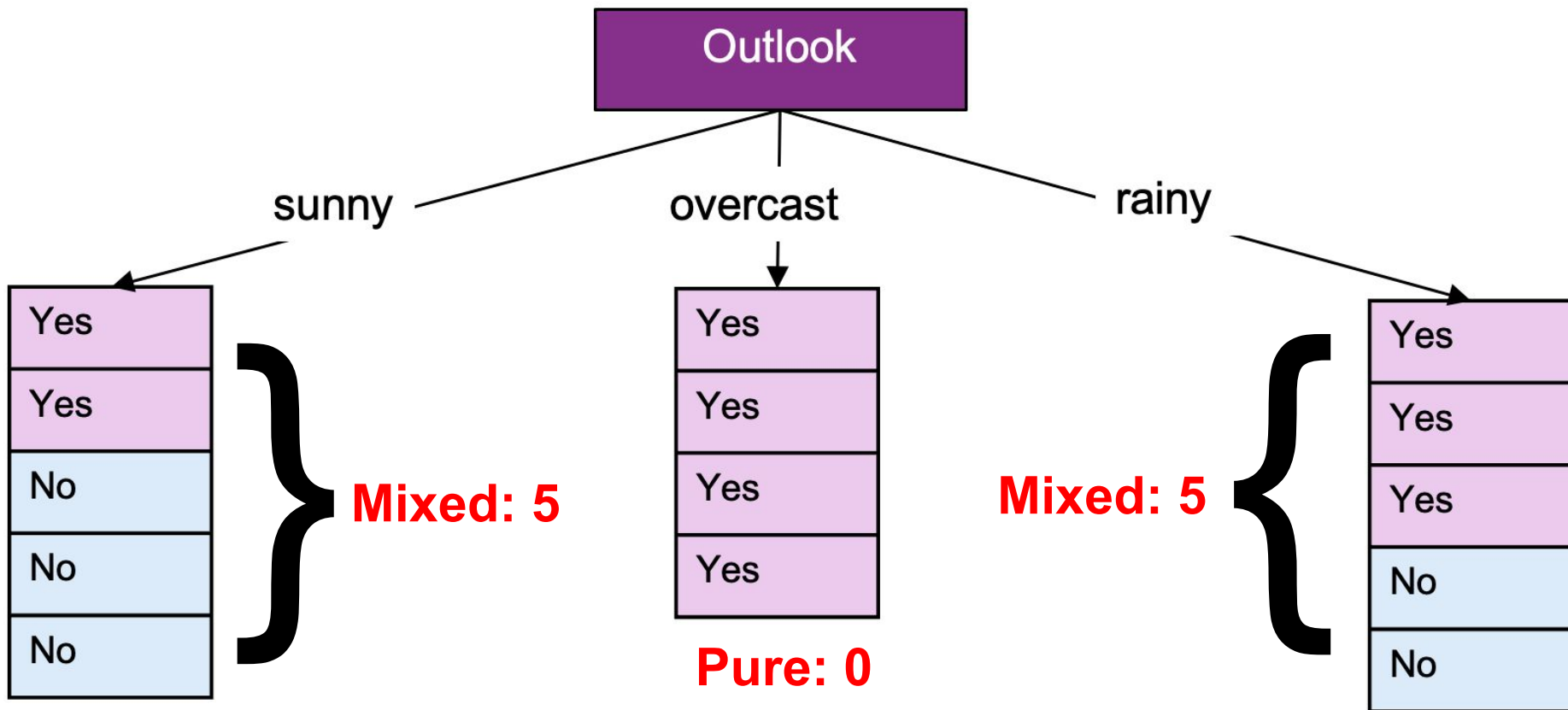
C. 10

D. 14

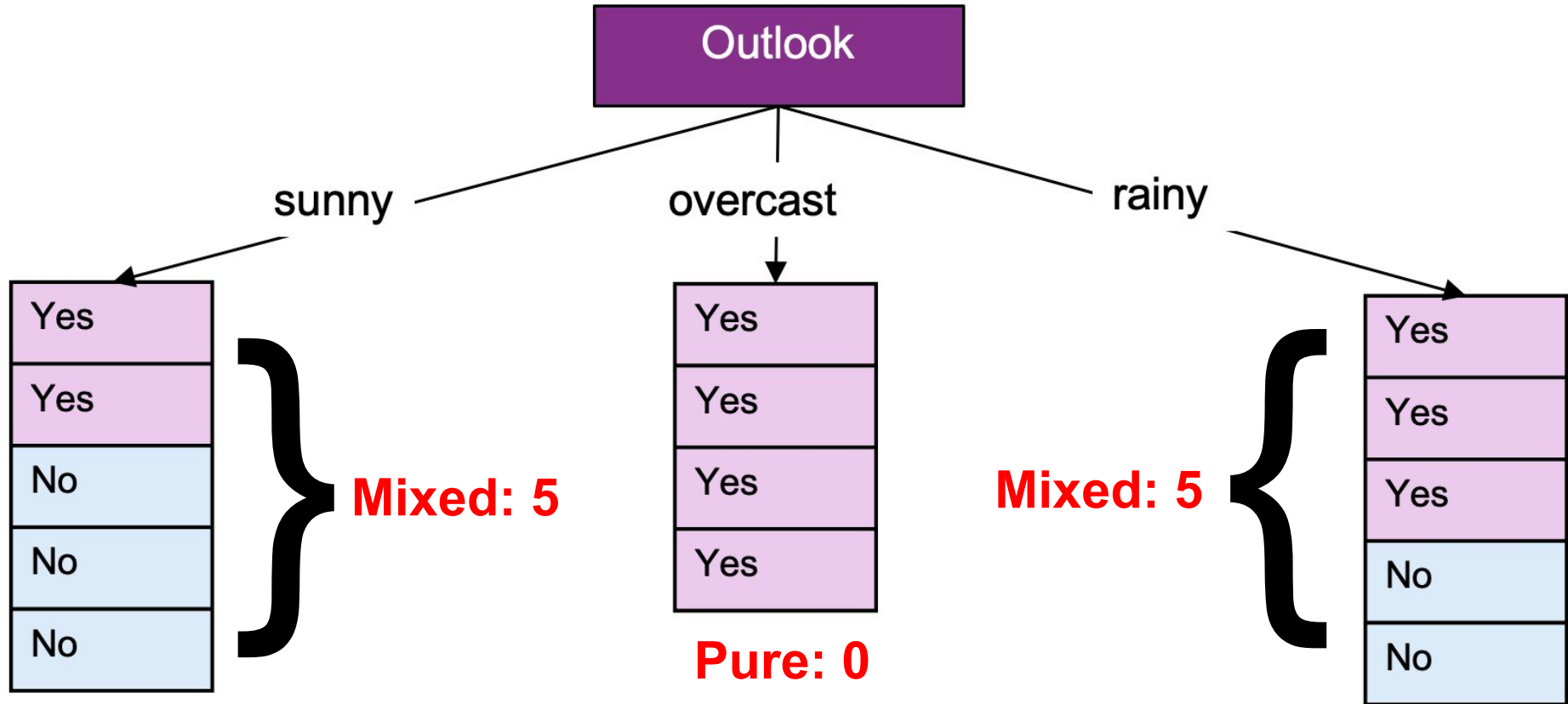
E. None of the above

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Q: What's the entropy if you split on Outlook?



$$\text{Overall entropy} = 5 + 0 + 5 = 10$$



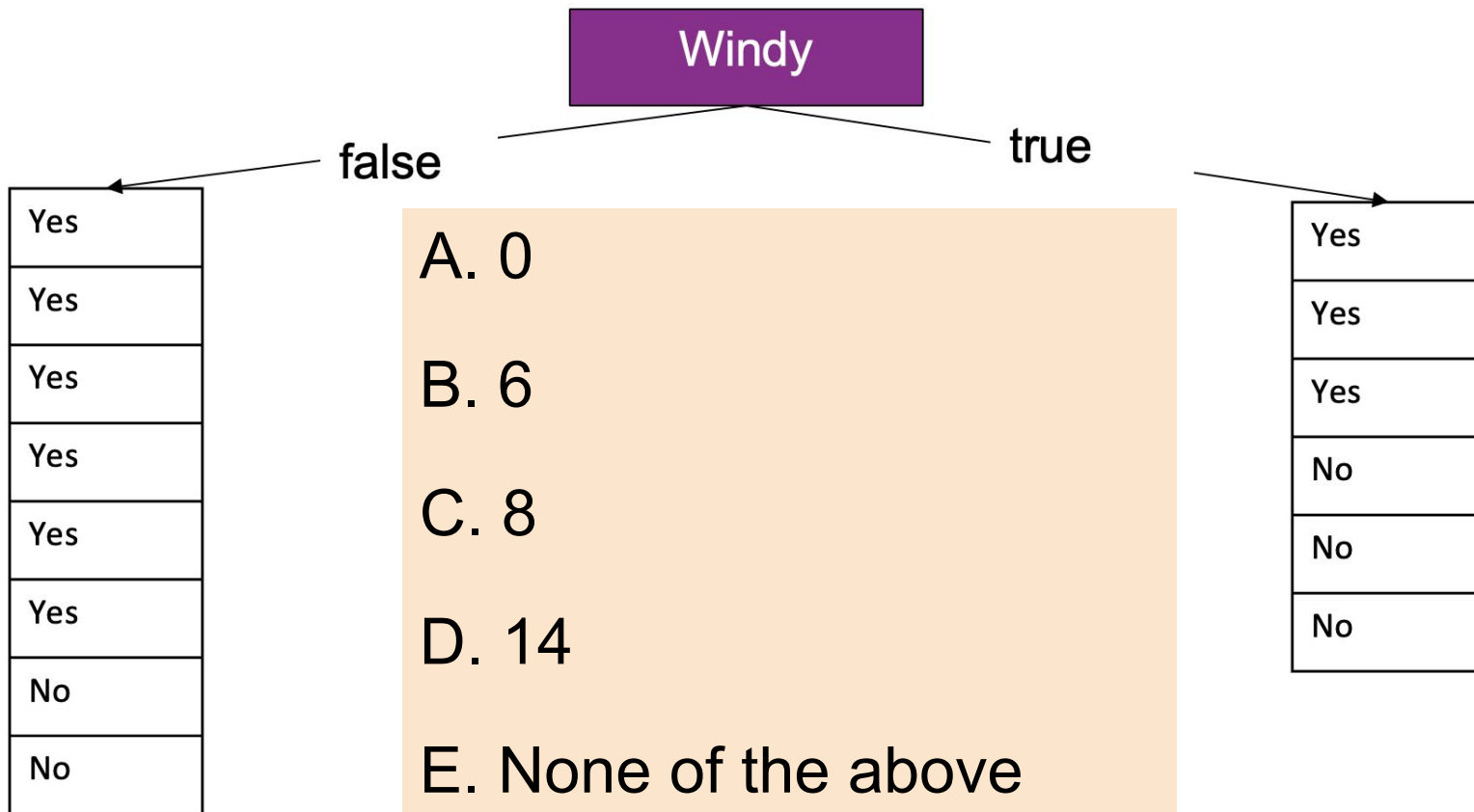
What's the entropy if you split on Windy?

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No



Clicker Question

Q: What's the entropy if you split on Windy?

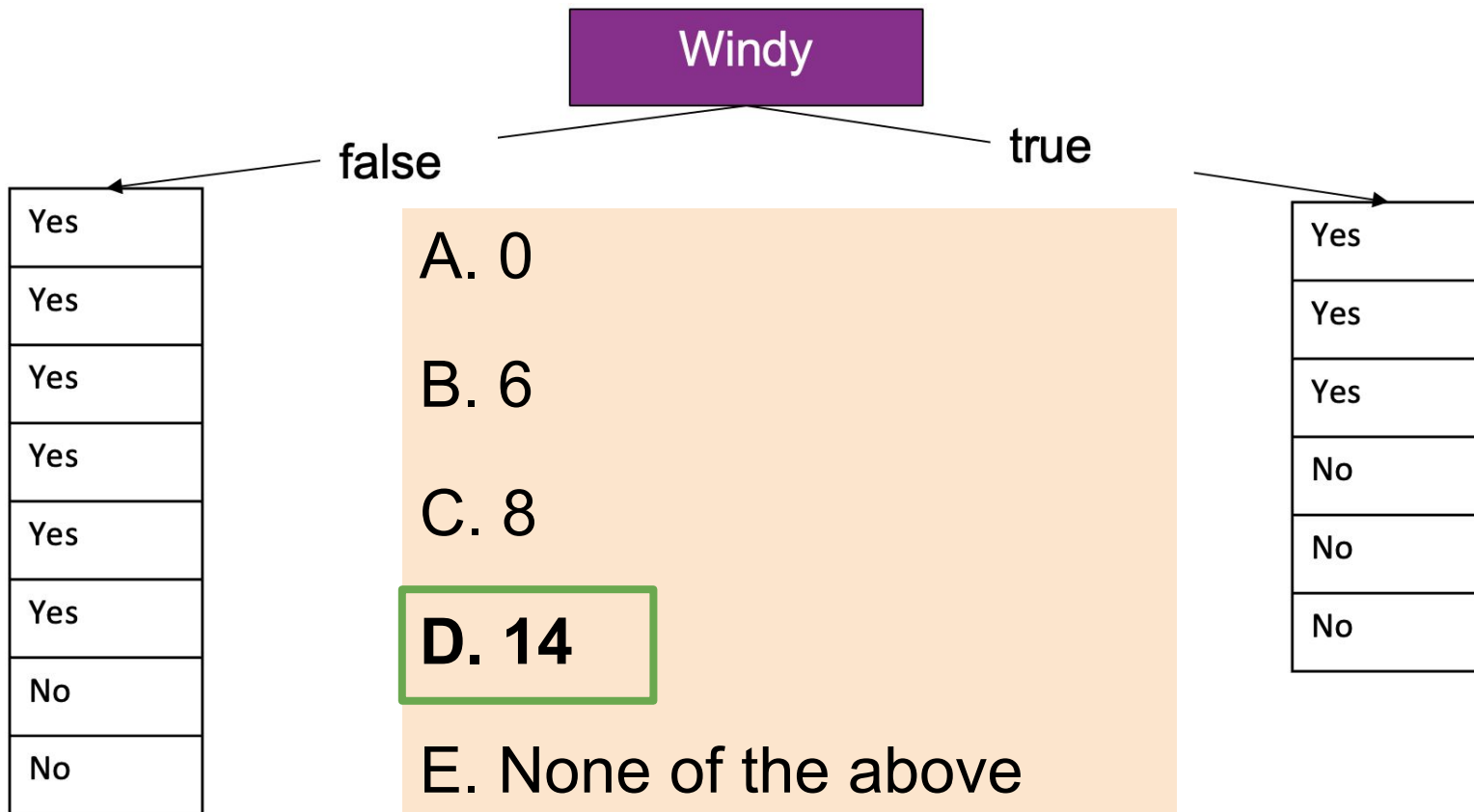


Clicker Question

Q: What's the entropy if you split on Windy?



iClicker



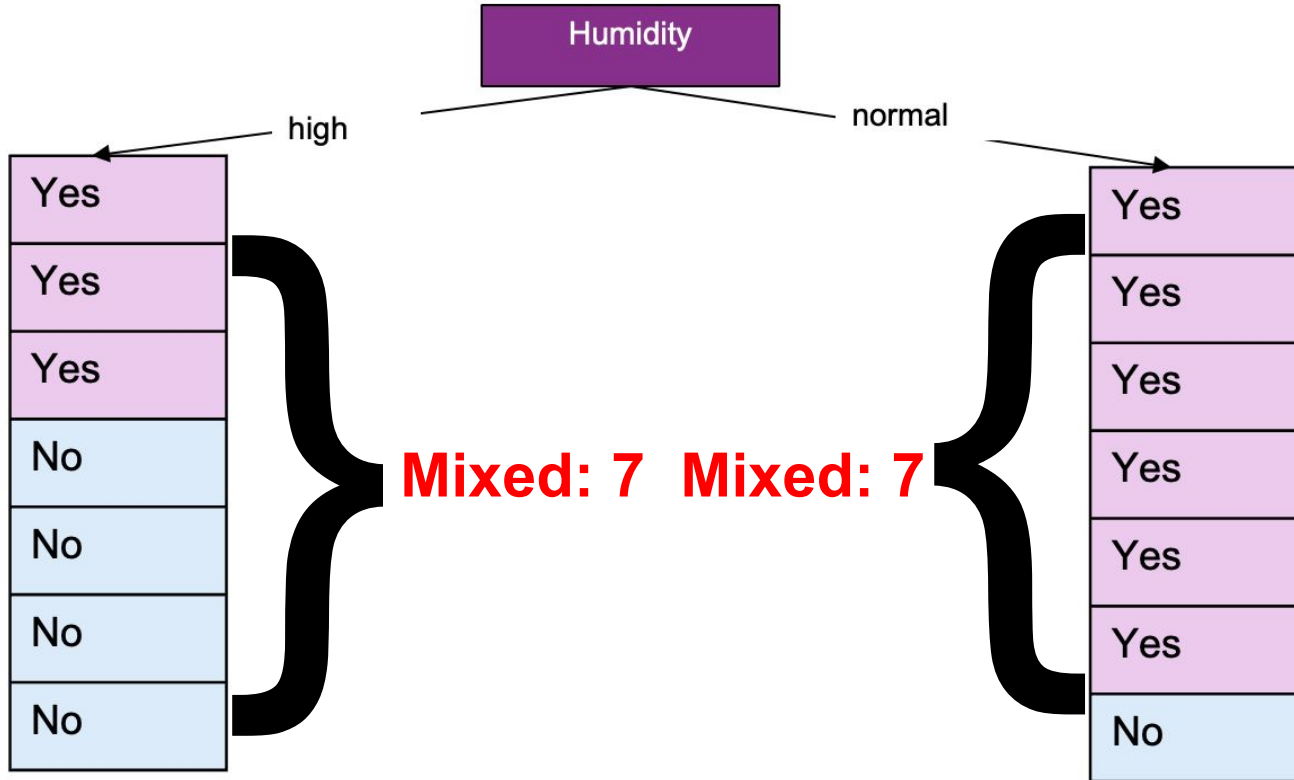


What's the entropy if you split on Humidity?

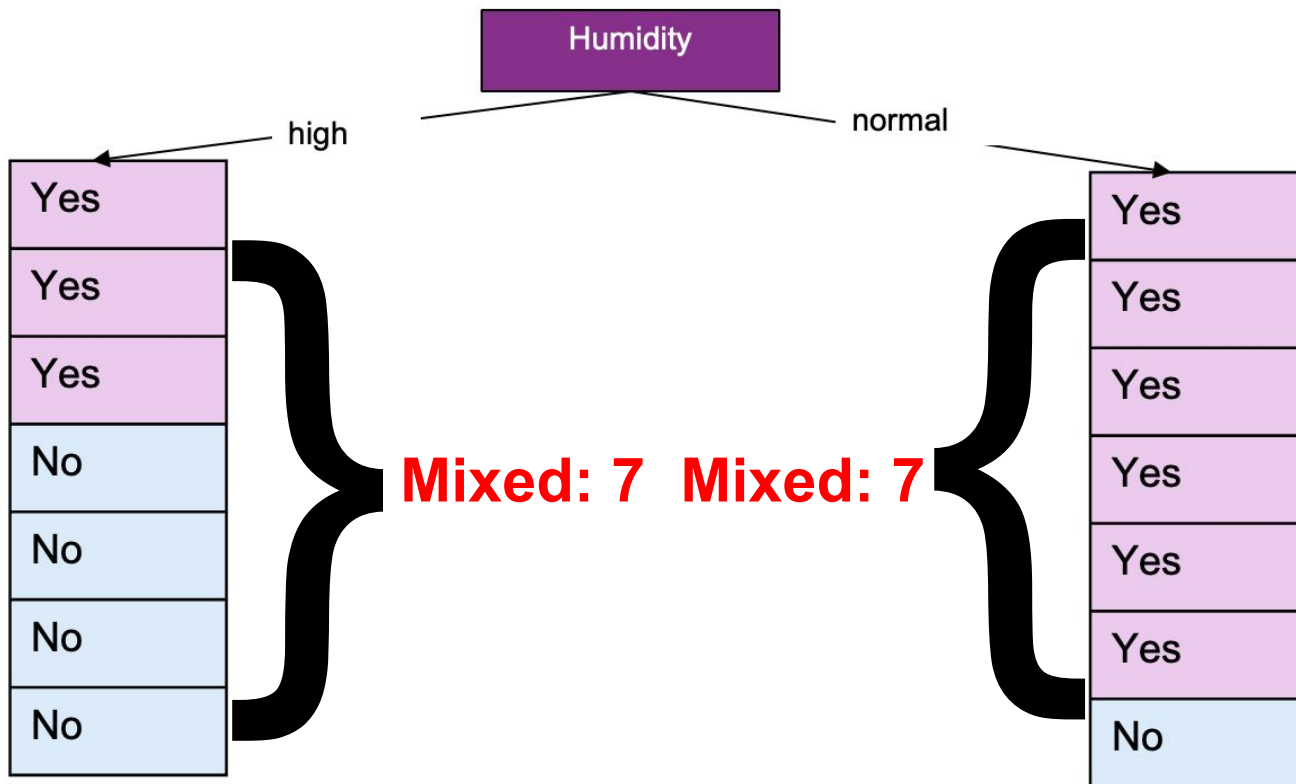
Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No



Q: What's the entropy if you split on Humidity?



Overall entropy = 7 + 7 = 14



Debrief



What is the best attribute to split on?

- Entropy if we split on Temperature = 14
- Entropy if we split on Outlook = 10
- Entropy if we split on Windy = 14
- Entropy if we split on Humidity = 14

Why?

What is the best attribute to split on?

- Entropy if we split on Temperature = 14
- **Entropy if we split on Outlook = 10**
- Entropy if we split on Windy = 14
- Entropy if we split on Humidity = 14

Why? It does the best job of **minimizing** entropy

Wrap up

CPSC 100

Course Project



CPSC 100 Course Project





Class Activity

- Today we will make groups for the project!
- I will help guide this process a bit in class...



Considerations for Group Formation

You will be doing this project in teams of 3 or 4, with no exceptions, as all of the milestones and details are designed for that many students.

As you can imagine, there is a fair bit of anxiety associated with group projects but know that this particular group project has several features that should make it possible to deal with any unforeseen circumstances.

There is a little bit of additional work needed to form a team, so we can make sure that all team members are on the same page, and teamwork can happen harmoniously.



Considerations for Group Formation

Considerations when looking for a team

It may be very tempting to do this, but try to avoid making a group with people you already know and have worked with in the past. I know this will be really hard and counter-intuitive, but group dynamics are very tricky and in my experience people that don't know each other from before tend to approach the project and the code of conduct differently.

Consideration 1: Interest in the topic

As I've said, the **most important** thing you should look for in a team member, is an interest in the topic of whatever dataset you end up choosing.



Considerations for Group Formation

Consideration 2: Diversity of knowledge, experience, and skills

Here is a self-assessment form that each members of your group should do.

Traits	Beginner	Novice	Expert
General familiarity with technology			
Presentations and communication			
Organization and Logistics			
Design/Aesthetic skills			
Attention to Detail			

The ideal group would have a healthy mix of traits in each of the group members. It's very unlikely that one person will be an "expert" in everything, and we want to make sure the group experience is such that everyone can learn and benefit in different ways. We are not going to police your groups, but the groups that I "worry about" the most, are those where all members declare themselves as "experts".



Considerations for Group Formation

Consideration 3: Availability

You should work with group members with whom you can easily find a weekly timeslot to meet, ideally in person and on campus. There is nothing more frustrating than group members who can't find a time to work on things simultaneously. If you are a morning person, try to find others who want to work in the mornings. If you stay up till 3 AM every night, it will likely not make a good match to work with team members who want to meet at 8 AM.



Considerations for Group Formation

Consideration 4: Goals for the project

You should work with group members with shared goals on what your final objective is. Your objective could be as simple as “do the bare minimum to pass the project”, or “do as much as I can given my heavy course and work load” (make sure to define what “as much as I can” means), or “try everything, and aim to do the best I can”, or “my goal is to make something I’m proud of that I can show my future employers”. Consider all those options above and make sure your team members are on the same page.

Q: Which topic are you most interested in (not finalized) ?



- A. Generative AI and student learning
- B. Technological inequality
- C. Algorithmic Bias
- D. Financial and Environmental impacts of technology
- E. Technology and Crime



Participation Question

What grade do you (tentatively) intend to contract for?

A. A+

B. A

C. B

D. C





Considerations for Group Formation

- . Share contact details with each other, and arrange for an initial meeting (Zoom, phone, instagram live, whatever y'all use these days).
 - o I will give you some class time to make the initial contact and exchange information.
- . In your initial meeting discuss the following items:
 - o What are your interests and goals with this project (datasets, research questions, what you want to learn)?
 - o How much time will you be able to dedicate to this project?
 - o How frequently will you meet or discuss the project?
 - o What are each of your strengths and weaknesses?
 - o What will you do if there is a conflict or if issues arises within the team?
 - o Do you all agree to stay in touch, and keep everyone informed despite how busy you are? I suggest setting a response time, say a response time of ~ 12-24 hours.
 - o Do you think you will be able to work together for this project?
 - o I suggest looking over Milestone 1 as a team and coming up with your answers together.
- . Once your initial meeting is done and you have answered those questions together, **one person in the team should submit Milestone 1 on Prairielearn** and answer all the questions.